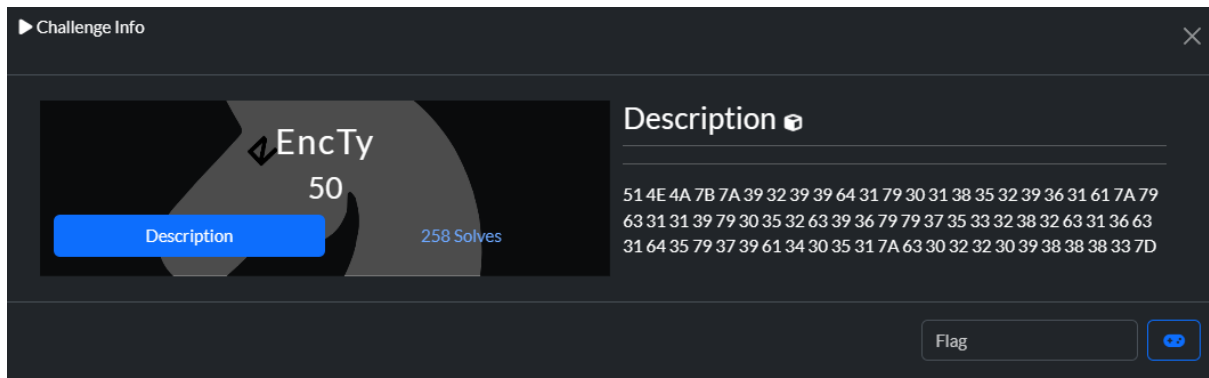


Challenge Name: EncTy - 50 (CRYPTO)

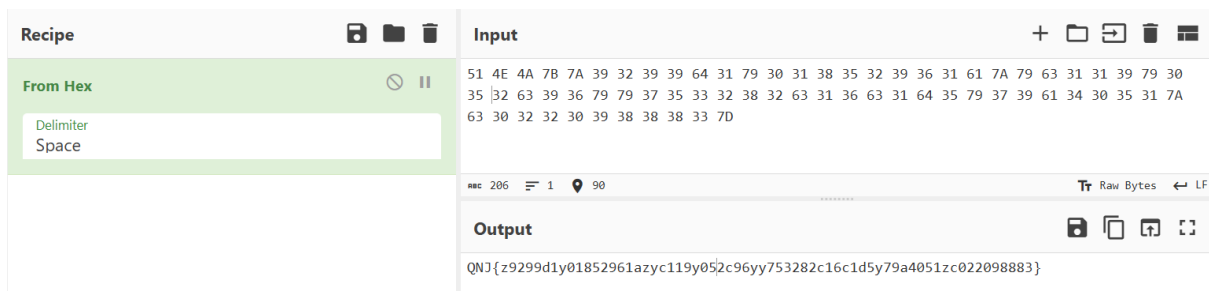


We were provided with the following hex string:

```
51 4E 4A 7B 7A 39 32 39 39 64 31 79 30 31 38 35 32 39 36 31 61 7A 79 63 31
31 39 79 30 35 32 63 39 36 79 79 37 35 33 32 38 32 63 31 36 63 31 64 35 79
37 39 61 34 30 35 31 7A 63 30 32 32 30 39 38 38 33 7D
```

Step 1 – Convert Hex to ASCII:

I first converted the hexadecimal string to ASCII using [CyberChef.org](https://cyberchef.org). This produced a string that looked like a flag format but was still encoded:



Step 2 – Recognize the Cipher:

Looking at the text, I noticed:

- Letters like **z**, **a**, **y**, **c** appeared repeatedly.
- Numbers remained unchanged.
- The letters seemed consistently shifted in the alphabet.

This pattern indicated a **Caesar cipher** was likely used.

Step 3 – Decode the Caesar Cipher:

I used a Caesar cipher decoder on dcode.fr to try different shifts. After decoding, the letters aligned correctly, revealing the final readable flag:



The screenshot shows the dCode.fr website interface. At the top, there's a banner about a new interface. Below it, the 'CAESAR CIPHER' section is active, with a breadcrumb trail: 'Cryptography > Substitution Cipher > Caesar Cipher'. The 'CAESAR CIPHER DECODER' section is visible, showing a search for 'boolean' and a list of results. The results table shows two entries: one for shift 24 and another for shift 21. The shift 24 result is highlighted, showing the ciphertext 'QNJ{z9299d1y01852961azyc119y052c96yy753282c16c1d5y79a4051zc022098883}'. The shift 21 result shows the plaintext 'VSO{e9299i1d01852961fedh119d052h96dd753282h16h1i5d79f4051eh022098883}'. The 'CAESAR CIPHER DECODER' section also shows the ciphertext 'QNJ{z9299d1y01852961azyc119y052c96yy753282c16c1d5y79a4051zc022098883}' and a button to 'DECRYPT (BRUTEFORCE)'.

Final:

The challenge combined **hexadecimal conversion** and **Caesar cipher decoding**. By first converting from hex to ASCII and then reversing the Caesar shift on letters, the flag was successfully recovered.

Flag: `SPL{b9299f1a01852961cbae119a052e96aa753282e16e1f5a79c4051be022098883}`

~By: 1nf1n1ty